



UNIVERSITY OF
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Statistical Sciences

DoSS Summer Bootcamp Probability Module 1

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Roadmap

A bridge connecting undergraduate probability and graduate probability

Undergraduate-level probability

- Concrete;
- Examples and scenarios;
- Rely on computation...

Roadmap

A bridge connecting undergraduate probability and graduate probability

Undergraduate-level probability

- Concrete;
- Examples and scenarios;
- Rely on computation...

Graduate-level probability

- Abstract (measure theory);
- Laws and properties;
- Rely on construction and inference...

Roadmap

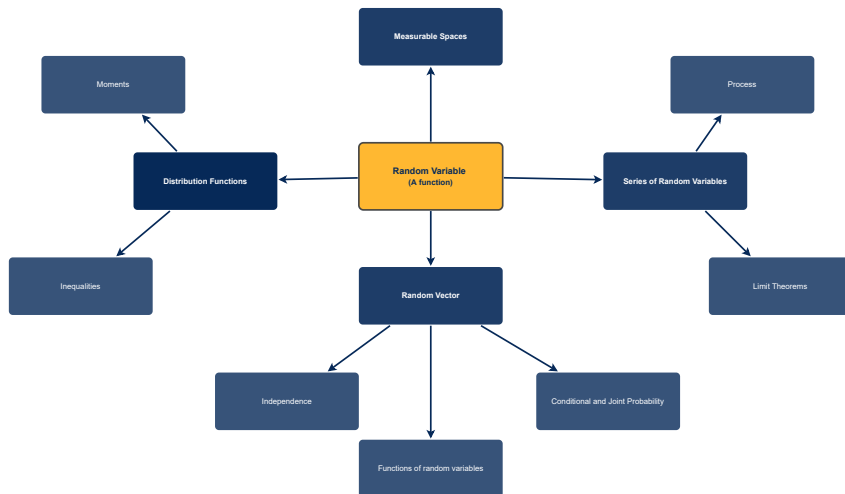


Figure: Roadmap

Outline

- Measurable spaces
 - ▷ Sample Space
 - ▷ σ -algebra
- Probability measures
 - ▷ Measures on σ -field
 - ▷ Basic results
- Conditional probability
 - ▷ Bayes' rule
 - ▷ Law of total probability

Measurable spaces

Sample Space

The sample space Ω is the set of all possible outcomes of an experiment.

Examples:

- Toss a coin: $\{H, T\} = \Omega$
- Roll a die: $\{1, 2, 3, 4, 5, 6\} = \Omega$

Measurable spaces

Sample Space

The sample space Ω is the set of all possible outcomes of an experiment.

Examples:

- Toss a coin: $\{H, T\}$
- Roll a die: $\{1, 2, 3, 4, 5, 6\}$

Event

An event is a collection of possible outcomes (subset of the sample space).

Examples:

- Get head when tossing a coin: $\{H\} \subset \Omega = \{H, T\}$
- Get an even number when rolling a die: $\{2, 4, 6\} \subset \Omega = \{1, 2, \dots, 6\}$

Discrete and Continuous Probability

Discrete random variable

For a discrete random variable X ,

$$\mathbb{P}(X \in A) = \sum_{x \in A} \mathbb{P}(X = x).$$

The probability of an event is obtained by adding point probabilities.

Continuous random variable

For a continuous random variable X with density f ,

$$\mathbb{P}(X \in A) = \int_A f(x) dx.$$

The probability of an event is obtained by integrating the density.

Why do we have two different definitions?

Discrete and Continuous Expectations

Discrete random variable

For a discrete random variable X ,

$$\mathbb{E}[X] = \sum_{x \in \mathbb{R}} x \mathbb{P}(X = x).$$

Continuous random variable

For a continuous random variable X with density f ,

$$\mathbb{E}[X] = \int_{\mathbb{R}} \underline{xf(x)} \, dx.$$

Again, a **sum** is used under discrete probability, while an **integral** is used under continuous probability.

Why Can't We Always Use Sums?

Suppose X is a continuous random variable. Then

$$\underline{\mathbb{P}(X = x) = 0} \quad \text{for every } x \in \mathbb{R}.$$

If we imitate discrete probability, then

$$1 = \mathbb{P}(X \in \mathbb{R}) \stackrel{?}{=} \sum_{x \in \mathbb{R}} \mathbb{P}(X = x) = \sum_{x \in \mathbb{R}} 0.$$

Real numbers

This would suggest

$$1 = 0,$$

which is an obvious contradiction.

Key point

The problem is not that each point has probability zero.

The problem is that we are trying to add uncountably many numbers.



Countability Is the Key

In ordinary calculus, infinite sums are defined for **countable** values:

$$\sum_{i=1}^{\infty} a_i.$$

countable sum

But an expression such as

$$\sum_{x \in [0,1]} a_x$$

is not a countable sum but an uncountable sum.

$[0,1]$ contains uncountably many values.

Motivation

Measure-theoretic probability is built around countable operations.
This motivates the definition of a σ -algebra.

A Countable Approximation of the Real Line

For each positive integer n , divide the real line into countably many intervals:

$$\left[\frac{i}{n}, \frac{i+1}{n} \right), \quad i \in \mathbb{Z}.$$

Then

$$\mathbb{R} = \bigcup_{i=-\infty}^{\infty} \left[\frac{i}{n}, \frac{i+1}{n} \right). \quad \leftarrow \text{union of countably many intervals.}$$

These intervals are disjoint, so countable additivity gives

$$\mathbb{P}(X \in \mathbb{R}) = \sum_{i=-\infty}^{\infty} \mathbb{P}\left(X \in \left[\frac{i}{n}, \frac{i+1}{n} \right)\right).$$

countable sum

Key point

Even though $\mathbb{R}$ is uncountable, it can be decomposed into countably many intervals.



Approximating Expectation by Countable Sums

Using the same intervals,

$$\left[\frac{i}{n}, \frac{i+1}{n} \right), \quad i \in \mathbb{Z},$$

we approximate X by the left endpoint of the interval containing it.

That is,

$$X \approx \frac{i}{n} \quad \text{when } X \in \left[\frac{i}{n}, \frac{i+1}{n} \right).$$

Therefore,

$$\mathbb{E}[X] \approx \sum_{i=-\infty}^{\infty} \frac{i}{n} \mathbb{P} \left(X \in \left[\frac{i}{n}, \frac{i+1}{n} \right) \right).$$

countable sum

Important observation

This formula uses only a **countable sum**, not an uncountable sum over individual points.



From Approximation to Measure-Theoretic Probability

As n becomes larger, the intervals

$$\left[\frac{i}{n}, \frac{i+1}{n} \right)$$

become smaller.

So the approximation should become more accurate

$$\mathbb{E}[X] \stackrel{?}{=} \lim_{n \rightarrow \infty} \sum_{i=-\infty}^{\infty} \frac{i}{n} \mathbb{P} \left(X \in \left[\frac{i}{n}, \frac{i+1}{n} \right) \right).$$

Key insights

These observations suggest that probabilities and expectations can be built from:

- probabilities of sets,
- countable unions of sets,
- countable sums.

Main idea

Measure-theoretic probability gives one unified language for both discrete and continuous probability by focusing on **sets** and **countable operations**.

Introduction to Measure Theory

σ -algebra

A σ -algebra (σ -field) $\mathcal{F}$ on Ω is a non-empty collection of subsets of Ω such that

- If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, $\mathcal{F}$ is closed under complement
 - If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$. $\mathcal{F}$ is closed under countable union
- countable operation

Remark: $\emptyset, \Omega \in \mathcal{F}$

Let $A \in \mathcal{F}$. Since $\mathcal{F}$ is closed under complement, $A^c \in \mathcal{F}$.

Since $\mathcal{F}$ is closed under countable union, $\underbrace{A \cup A^c}_{\Omega} = \mathcal{F}$.

Thus $\Omega \in \mathcal{F}$.

Again, taking the complement of Ω , we conclude $\emptyset = \Omega^c \in \mathcal{F}$.

Prop. $\mathcal{F}$ is closed under countable intersection:

$$\text{If } A_1, A_2, \dots \in \mathcal{F}, \quad \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}.$$

(pf.) First observe that

$$\left(\bigcap_{i=1}^{\infty} A_i \right)^c = \bigcup_{i=1}^{\infty} A_i^c$$

Since $\mathcal{F}$ is closed under complement, $A_i^c \in \mathcal{F}$ for all i .

Since $\mathcal{F}$ is closed under countable union, $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{F}$.

Since $\mathcal{F}$ is closed under complement,

$$\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i^c \right)^c \in \mathcal{F} //$$

Probability measures

algebra

Measures on σ -field

A function $\mu : \mathcal{F} \rightarrow R^+ \cup \{+\infty\}$ is called a measure if

- $\mu(\emptyset) = 0$,
- If $A_1, A_2, \dots \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$, then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

If $\mu(\Omega) = 1$, then μ is called a probability measure.

countable additivity

Probability measures

Measures on σ -field

A function $\mu : \mathcal{F} \rightarrow \mathbb{R}^+ \cup \{+\infty\}$ is called a measure if

- $\mu(\emptyset) = 0$,
- If $A_1, A_2, \dots \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$, then $\mu(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

If $\mu(\Omega) = 1$, then μ is called a probability measure.

Properties:

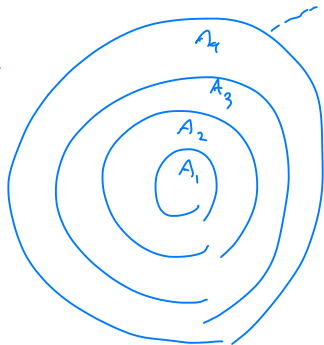
- Monotonicity: $A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$
- Subadditivity: $A \subseteq \cup_{i=1}^{\infty} A_i \Rightarrow \mu(A) \leq \sum_{i=1}^{\infty} \mu(A_i)$ *← do not assume A_i 's are disjoint.*
- Continuity from below: $A_i \nearrow A \Rightarrow \mu(A_i) \nearrow \mu(A)$
- Continuity from above: $A_i \searrow A$ and $\mu(A_i) < \infty \Rightarrow \mu(A_i) \searrow \mu(A)$

$A_1 \subset A_2 \subset A_3 \subset \dots, \bigcup_{i=1}^{\infty} A_i = A$

Probability measures

Proof of continuity from below:

Construct disjoint B_i by $B_i = A_i \setminus A_{i-1}$ $i \geq 2$ ✓
 $B_1 = A_1$



Since $A = \bigcup_{i=1}^{\infty} B_i$ and by countable additivity

$$\begin{aligned}\mu(A) &= \sum_{i=1}^{\infty} \mu(B_i) = \sum_{i=1}^{\infty} \underbrace{\mu(A_i \setminus A_{i-1})}_{= \mu(A_i) - \mu(A_{i-1})} \\ &= \lim_{n \rightarrow \infty} \underbrace{\sum_{i=1}^n (\mu(A_i) - \mu(A_{i-1}))}_{\uparrow \mu(A_n)} = \lim_{n \rightarrow \infty} \mu(A_n)\end{aligned}$$



Probability measures If $A_i \supseteq A$, $\mu(A_i) < \infty$, then $\lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$

Proof of continuity from above:

Define $B_i = A_i \setminus A$, then $B_i \cap A_i \setminus A \rightarrow$ we reduce to continuity from below.

By continuity from below, we have

$$\lim_{n \rightarrow \infty} \mu(B_n) = \mu(A_i \setminus A)$$

$$\Leftrightarrow \lim_{n \rightarrow \infty} \mu(A_i \setminus A_n) = \mu(A_i \setminus A)$$

Remark: $\mu(A_i) < \infty$ is vital. $\textcircled{1} \Leftrightarrow \lim_{n \rightarrow \infty} (\mu(A_i) - \mu(A_n)) = \mu(A_i) - \mu(A)$

$$\textcircled{2} \therefore \lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$$

$\textcircled{1} \mu(A \setminus B) = \mu(A) - \mu(B)$ holds
if $\mu(A) < \infty$ and $B \subset A$

$\textcircled{2} \infty - a = \infty - b$ is possible even if $a \neq b$

Probability measures

Examples:

$$\Omega = \{\omega_1, \omega_2, \dots\}, A = \{\omega_{a_1}, \dots, \omega_{a_i}, \dots\} \Rightarrow \mu(A) = \sum_{j=1}^{\infty} \mu(\omega_{a_j}).$$

Therefore, we only need to define $\mu(\omega_j) = p_j \geq 0$.

If further $\sum_{i=1}^{\infty} p_j = 1$, then μ is a probability measure.

- Toss a coin:

$$\Omega = \{H, T\}, \quad \mu(H) = p_H, \quad \mu(T) = p_T \quad \text{where} \quad p_H + p_T = 1,$$

Then μ is a probability measure on Ω .

- Roll a die:

$$\Omega = \{1, 2, \dots, 6\}, \quad \mu(1) = p_1, \quad \dots, \quad \mu(6) = p_6, \quad \sum_{i=1}^6 p_i = 1$$

Then μ is a probability measure.

Example: Tossing a Coin Twice

Consider the sample space $\Omega = \{HH, HT, TH, TT\}$.

One possible σ -algebra is the power set

$$2^{\Omega},$$

which contains all $2^4 = 16$ subsets of Ω .

Another example is

$$\mathcal{F} = \{\emptyset, \Omega, \{HH, HT\}, \{TH, TT\}\}.$$


complement

This collection satisfies

- $\emptyset, \Omega \in \mathcal{F}$,
- complements remain in $\mathcal{F}$,
- countable unions remain in $\mathcal{F}$.

The Smallest and Largest σ -Algebras

For any sample space Ω , there are always two extreme examples.

The trivial σ -algebra

$$\{\emptyset, \Omega\}.$$

Only two events are measurable.

The power set

$$2^\Omega = \{A : A \subseteq \Omega\}.$$

Every subset of Ω is measurable.

Observation

Every σ -algebra lies between these two extremes:

$$\{\emptyset, \Omega\} \subseteq \mathcal{F} \subseteq 2^\Omega.$$

Generating a σ -Algebra

Instead of specifying every measurable set, we often begin with a collection of subsets.
Suppose

$$\mathcal{C} = \{A_1, A_2, \dots\}.$$

We then define $\sigma(\mathcal{C})$ to be the **smallest σ -algebra containing every set in $\mathcal{C}$.**

Intuitively, $\sigma(\mathcal{C})$

- include every set in $\mathcal{C}$,
- include all complements,
- include all countable unions,
- repeat until no new sets are created.

This construction allows us to build useful σ -algebras from simple building blocks.

Example: The Borel σ -Algebra → necessary for continuous probability

On the real line, the most important σ -algebra is the **Borel σ -algebra**.
It is generated by all open intervals:

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) : a < b\}).$$

Therefore,

$$(a, b) \in \mathcal{B}(\mathbb{R})$$

for every open interval, and so are all sets obtained by repeatedly taking

- complements,
- countable unions,
- countable intersections.

Examples of Borel sets

Examples of Borel sets include

$$\underbrace{[a, b]}, \quad \underbrace{(a, b]}, \quad \underbrace{(-\infty, a]}, \quad \underbrace{\mathbb{Z}}, \quad \underbrace{\mathbb{Q}}, \quad \{a\}$$

$[a, b] = \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, b + \frac{1}{n})$

Conditional probability

Original problem:

- What is the probability of some event A ?
- $P(A)$ is determined by our probability measure.

New problem:

- Given that B happens, what is the probability of some event A ?
- $P(A | B)$ is the conditional probability of the event A given B .

Conditional probability

Original problem:

- What is the probability of some event A ?
- $P(A)$ is determined by our probability measure.

New problem:

- Given that B happens, what is the probability of some event A ?
- $P(A \mid B)$ is the conditional probability of the event A given B .

Example:

- Roll a die: $P(\{2\} \mid \text{even number})$

Conditional probability

Definition

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Remark: Does conditional probability $P(\cdot | B)$ satisfy the axioms of a probability measure?

$$1) \quad P(\emptyset | B) = \frac{P(\emptyset \cap B)}{P(B)} = \frac{P(\emptyset)}{P(B)} = \frac{0}{P(B)} = 0. \quad \text{countable additivity}$$

$$2) \text{ If } A_1, A_2, \dots \in \mathcal{F}, \quad A_i \cap A_j = \emptyset \text{ if } i \neq j, \\ P\left(\bigcup_{i=1}^{\infty} A_i | B\right) \stackrel{\text{definition}}{=} \frac{P\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{P(B)} \stackrel{\text{countable additivity}}{=} \frac{\sum_{i=1}^{\infty} P(A_i \cap B)}{P(B)} \\ \stackrel{\text{definition}}{=} \sum_{i=1}^{\infty} P(A_i | B)$$

$$3) \quad P(Q|B) = \frac{P(Q \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

Conditional probability

Multiplication rule

$$P(A \cap B) = \underbrace{P(A | B)}_{\substack{\text{"} \\ P(A \cap B) \\ P(B)}} P(B) = \underbrace{P(B | A)}_{\substack{P(B \cap A) \\ P(A)}} P(A)$$

Generalization:

Law of total probability

Let $A_1, A_2, \dots, A_n$ be a partition of Ω , such that $P(A_i) > 0$, then

$$P(B) = \sum_{i=1}^n P(A_i) P(B | A_i)$$

cf.)
$$P(B) = \sum_{i=1}^n P(B \cap A_i) \stackrel{\text{multiplication rule}}{=} \sum_{i=1}^n P(A_i) P(B | A_i)$$

Problem Set

Problem 1: Prove that for a σ -algebra $\mathcal{F}$, if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$.

Problem 2: Prove monotonicity and subadditivity of measure μ on σ -algebra.

Problem 3: (Monty Hall problem) Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

(Assumptions: the host will not open the door we picked and the host will only open the door which has a goat.)